

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 2	4 May 2022
Time : 1.5 hours	Maximum marks: 52
	Full marks: 50

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	TOTAL
3	6	3	5	5	6	10	9	5	52

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Section A

In this section, give your final answer only. Do not show your working.

Question 1. [3 points] Let A be a square matrix such that $\det(A) = 5$ and $\det(2A) = 40$. Find

(a) the size of A ;

--

(b) $\det(A^{-1})$;

--

(c) $\det(A^T A)$.

--

Question 2. [6 points] Let $A = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

--

(b) Find all the eigenvectors corresponding to the largest eigenvalue of A .

--

Question 3. [3 points] Write down a basis for the vector space of all symmetric 3×3 matrices.

--

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Question 4. [5 points] Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 4 \\ 2 & -4 & 3 & 4 & 9 \\ 3 & -6 & 4 & 5 & 13 \end{pmatrix}$ and $U = \begin{pmatrix} 1 & -2 & 1 & 1 & 4 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$;

U is a row-echelon form of A (you need not check this).

(a) Find the nullity and rank of A .

--

(b) Find the nullity of A^T .

--

(c) Write down a basis for (i) $\text{CS}(A)$ (ii) $\text{RS}(A)$.

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Section B

In this section, give full answers to all the questions. Show all your working.

Question 5. [5 points] Let V be the vector space of all $n \times n$ matrices and

$$S = \{A \in V : \det(A) = 0\}.$$

Answer the following questions, giving reasons for all your answers.

(a) Is S closed under scalar multiplication?

--

(b) Is S closed under addition?

--

(c) Is S a subspace of V ?

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 6. [6 points] The function $S : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ is defined by $S(A) = \frac{1}{2}(A + A^T)$.

(a) Show that S is a linear transformation.

(b) Find the kernel of S .

(c) Find the range of S

Student Number:

--	--	--	--	--	--	--	--	--

Question 7. [10 points] Let $S = \{x^3 - x, x^3 + 1, x + 1, x^2\}$.

(a) Is S a linearly independent subset of P_3 ? Give a reason for your answer.

--

(b) Explain why S cannot span P_3 .

--

(c) A polynomial $p(x) = ax^3 + bx^2 + cx + d$ will be in $\text{span}(S)$ if and only if the coefficients a, b, c, d satisfy a certain equation. Find this equation.

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

- (d) Find a subset B of S that is a basis for $\text{span}(S)$, and explain why it is a basis.

- (e) Is it possible to find a polynomial $p(x) \in P_3$ such that $S \cup \{p(x)\}$ is a basis for P_3 ? Give a reason for your answer.

- (f) Is it possible to find a polynomial $p(x) \in P_3$ such that $B \cup \{p(x)\}$ is a basis for P_3 , where B is the set you found in (d). Give a reason for your answer. If your answer is “yes”, give such a $p(x)$.

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

(d) Carefully state the result proved here.

(e) Using the notation of the proof, give the coordinate vector $[\mathbf{v}]_B$ of $\mathbf{v}$ with respect to the basis B . Why is the result proved here needed to define the concept of a coordinate vector?

(f) Prove that for any $\mathbf{v}, \mathbf{w} \in V$, $[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 9. [5 points] Let V be a finite dimensional vector space and U be a subspace of V . Show that there is a linear transformation $T : V \rightarrow V$ such that $\ker(T) = U$ and $T(T(\mathbf{v})) = T(\mathbf{v})$ for all $\mathbf{v} \in V$. [Hint: Consider a basis for U . Is there a basis for V containing this basis for U ?]